A. Euclid, Sequence and Two Numbers

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: math, number theory, sortings | https://codeforces.com/problemset/problem/2234/A

We define a Euclid algorithm sequence of length k ($k \geq 2$) for two positive integers $x \geq y$ as the following sequence of positive integers:

- $a_1, a_2, \ldots, a_k$, where $a_1 = x$, $a_2 = y$, and for any i ($1 \leq i \leq k - 2$), the equality $a_{i+2} = (a_i \bmod a_{i+1})$ holds*.

For example, for $x = 13$, $y = 8$, $k = 4$, the corresponding Euclid algorithm sequence is $a = [13, 8, 5, 3]$. ($a_3 = 13 \bmod 8 = 5$, $a_4 = 8 \bmod 5 = 3$).

You are given a sequence $b_1, b_2, \ldots, b_n$. You need to determine whether it is possible to permute its elements so that it becomes a Euclid algorithm sequence for some two positive integers $x \geq y$.

*$x \bmod y$ denotes the remainder when x is divided by y.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 500$). The description of the test cases follows.

The first line of each test case contains an integer n ($2 \leq n \leq 100$) — the size of the sequence.

The second line of each test case contains n integers $b_1, b_2, \ldots, b_n$ ($1 \leq b_i \leq 10^9$) — the sequence b.

Output

For each test case, if it is possible to permute the elements of sequence b so that there exists a suitable pair of positive integers $x \geq y$, output x, y on a separate line. Otherwise, output -1 on a separate line.

If there are several suitable pairs x, y, you may output any of them.

Example

Input

```
6
2
1 1
2
1 2
4
1 2 3 4
3
6 4 2
4
3 8 13 5
3
1 1 1
```

Output

```
1 1
2 1
-1
6 4
13 8
-1
```

Note

In the first test case, the pair (1, 1) is suitable: for $x = 1$, $y = 1$, $k = 2$, $a_1 = x = 1$, $a_2 = y = 1$, and the sequence $a = [1, 1] = b$ is obtained.

In the third test case, it can be shown that no suitable pair (x, y) exists.

In the fourth test case, the pair (6, 4) is suitable: for $x = 6$, $y = 4$, $k = 3$, $a_1 = x = 6$, $a_2 = y = 4$, $a_3 = (a_1 \bmod a_2) = (6 \bmod 4) = 2$, and the sequence $a = [6, 4, 2] = b$ is obtained.